

percent in more than a decade of trading through many volatile markets. The key to this amazing feat of risk management has been the firm's exposure reduction rules. BlueCrest's CEO, Michael Platt, restricts himself and other discretionary traders at the firm to a loss limit of 3 percent (from the starting allocation) before the exposure allocation is cut in half. A loss of another 3 percent leads to the removal of the trader's entire allocation. These rigid controls severely limit the loss any trader can realize from a starting allocation. The rules encourage traders to be extremely conservative in their risk-taking at the start of each year. As traders register gains, however, they can increase their risk levels because the original exposure reduction thresholds remain unchanged for the year. In this manner, upside potential is open-ended, while downside risk is severely curtailed. Barring huge overnight gaps in the market, larger losses can only occur through the surrender of year-to-date profits rather than losses of original capital. Benedict utilizes a similar risk management philosophy. Any time he approaches a 2.5 percent loss in any given month, he significantly reduces net exposure and continues to trade in smaller size until the loss is recovered. In this manner, he severely constrains his potential loss in any given month.

- **Position size adjustments for changes in volatility**—As examples of this approach, in 2008, both Woodruff and Clark cut their exposure levels by approximately a factor of four in response to the steep increase in volatility.
- **Trade-dependent risk controls**—Some trades are inherently risk-constrained, whereas other trades have open-ended risk. In recognition of these differences, uniform risk controls across all trades may not be appropriate. For example, when Thorp implemented arbitrage trades that had a well-defined maximum theoretical risk, he did not consider reducing exposure if the position went against him. In contrast, when he employed a trend-following strategy, in which the trades were directional and the risk was open-ended, he made exposure reduction on drawdowns part of the methodology.